🧠 What is JSON?

JSON (JavaScript Object Notation) is a lightweight data format used for **storing and exchanging data** between a server and a client (like in APIs).
It is human-readable and easy to work with in Python.

🧩 JSON Format Example

A typical JSON looks like this:

```
{
  "name": "John",
  "age": 30,
  "isEmployed": true,
  "skills": ["Python", "Django", "Machine Learning"],
  "address": {
    "city": "New York",
    "zip": "10001"
  }
}
```

🐍 Python and JSON

Python provides a built-in module called **json** to work with JSON data.

👉 Import the Module

```
import json
```

🔄 JSON <-> Python Data Types Conversion

JSON Type	Python Equivalent
object	dict
array	list
string	str
number	int / float
true	True
false	False
null	None

1️⃣ Converting JSON String → Python Object

Use **json.loads()** (“load string”)

```
import json


# JSON string
json_data = '{"name": "Alice", "age": 25, "isStudent": false}'


# Convert to Python dictionary
python_data = json.loads(json_data)


print(python_data)
print(type(python_data))  # dict
```

Output:

```
{'name': 'Alice', 'age': 25, 'isStudent': False}
<class 'dict'>
```

2️⃣ Converting Python Object → JSON String

Use **json.dumps()** (“dump string”)

```
import json


# Python dictionary
data = {
    "name": "Bob",
    "age": 28,
    "skills": ["C#", "Python"]
}


# Convert to JSON string
json_string = json.dumps(data)


print(json_string)
print(type(json_string))  # str
```

Output:

```
{"name": "Bob", "age": 28, "skills": ["C#", "Python"]}
<class 'str'>
```

3️⃣ Formatting JSON Output (Pretty Print)

You can make JSON output **more readable** using indent and sort_keys.

```
print(json.dumps(data, indent=4, sort_keys=True))
```

Output:

```
{
    "age": 28,
```

```
  "name": "Bob",
  "skills": [
    "C#",
    "Python"
  ]
}
```

4️⃣ Working with JSON Files

✅ Read from a JSON file

```
import json

with open('data.json', 'r') as file:
  data = json.load(file)  # read + convert to dict
  print(data)
```

✅ Write to a JSON file

```
import json

data = {
  "name": "Charlie",
  "age": 35,
  "city": "Delhi"
}

with open('data.json', 'w') as file:
  json.dump(data, file, indent=4)
```

🧰 Practical Example — API Response Simulation

Imagine you received an API response as JSON:

```
import json

response = '''
{
  "id": 101,
  "title": "Python JSON Tutorial",
  "tags": ["python", "json", "tutorial"],
  "published": true
}
'''

# Convert to Python object
post = json.loads(response)

# Access data
print(post["title"])
print(post["tags"][1])
```

Output:

```
Python JSON Tutorial
json
```

⚠️ Common Errors

Error	Cause	Fix
JSONDecodeError	Invalid JSON format	Ensure double quotes, no trailing commas
TypeError: Object of type X is not JSON serializable	Trying to dump non-supported types (e.g., datetime)	Convert to string before dumping

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🧩 Summary

Task	Function
Parse JSON string → Python object	json.loads()
Convert Python object → JSON string	json.dumps()
Read JSON file → Python object	json.load()
Write Python object → JSON file	json.dump()